ASTROVERSE · INDIVIDUAL PROJECT · 4 NOVEMBER 2026

# Reaching space. Understanding orbit.

Mission Forge connects NASA data, a complete flight, and an explanation a learner can test.

**Shashank Pandey**
Browser-based spacecraft construction and orbital learning

01 · THE PROBLEM AND THE SOLUTION

# Altitude alone does not make an orbit.

A learner should be able to explain why a rocket falls back—and what to change next.

Predict → Build & fly → Inspect evidence → Reflect & retry

### Retain the prediction

Record an explanation and a speed estimate before the showcase. Wrong predictions remain visible.

### Explain the actual flight

Compare periapsis and recorded telemetry with the ideal reference. Save a reflection and export the learning record.

> Educational effectiveness is a hypothesis to test with learners; no learning-gain study is claimed.

02 · NASA DATA THAT CHANGES THE SIMULATION

# From a source field to a flight decision.

**12**

### Body radii

JPL/NAIF pck00011 triaxial axes converted into volume-equivalent spheres.

**10**

### Gravity parameters

NASA GMAT defaults converted from $km^3/s^2$ to $m^3/s^2$. Europa/Titan gravity is separately labelled.

**178**

### Earth/Mars density bands

Eight preserved NASA data/method snapshots, including EGM96 gravity, plus 12 signed NAIF rotation periods, with pinned provenance and checksums.

Radius controls altitude and the surface boundary. μ controls gravity and orbital speed. The in-game panel exposes both values and their source fields.

Verified channel: NASA's official nasa/GMAT repository. Archived defaults, not live ephemerides or NASA endorsement.

03 · SCIENCE WITH INSPECTABLE LIMITS

# Calculate independently. Then test the integrator.

## 7,788 m/s

**Earth · 200 km**

Ideal circular speed $v = \sqrt{(\mu/r)}$

Period $T = 2\pi\sqrt{(r^3/\mu)}$

## 1,634 m/s

**Moon · 100 km**

A different gravity well changes the flight requirement. The same formula provides a useful comparison.

**Independent references:** Decimal equations, actual Orekit trajectories and gravity vectors, native-C atmosphere cases, and a published ERFA frame matrix. Inputs, forces and tolerances are exposed alongside results.

> These are ideal two-body numerical checks. They do not validate all atmospheres, ephemerides, re-entry conditions, or learning outcomes.

04 · IMPLEMENTATION AND RELIABILITY

# A complete mission, across touch and keyboard.

### Build & fly

Four scored missions, solar exploration, local fleets/docking, programmable controls and custom worlds.

### Accessible controls

WASD defaults, remapping, independent touch steering/throttle, stage and RCS buttons, mirrored layouts and prepared offline play.

### Reproducible evidence

309 regression checks; 12 campaign validations; 51 external-reference checks; replay, verified local backups, source checksums and complete browser showcase checks.

NASA gravity changes exposed an overly long lunar-return solution. The planner now checks return time, and the crewed sequence is validated through recovery.

The stock Expert Lunar Orbiter remains a budget challenge with expected partial success. Mobile tests are emulated; physical-device testing remains.

05 · LIVE SHOWCASE

# One question. One dependable flight.

1. **Show NASA inputs:** compare Earth and the Moon and run the science checks.
2. **Save a prediction:** what keeps the spacecraft in orbit, and how fast must it travel?
3. **Launch First Orbit:** use the tested starter, guidance, and ×10 time speed.
4. **Inspect periapsis:** show the difference between reaching space and staying in orbit.
5. **Explain and export:** compare the recorded flight, save a reflection, and open the learning journal.

Open Mission Forge  · Presenter script  · Recorded example 

> If using a backup recording or exported result, identify it clearly. Gameplay uses bundled data and needs no live NASA API.

06 · IMPACT, EVIDENCE, AND NEXT STEPS

# Make the reasoning visible.

The contribution is a clear path from NASA data to a testable prediction, a flight observation, and a reasoned next decision.

### Demonstrated

Traceable inputs, working missions, numerical consistency checks, responsive controls, and an exportable learning loop.

### Next validation

Actual learner sessions, physical phones, another browser engine, and broader atmosphere/transfer accuracy checks.

Detailed project report  · Learner pilot protocol 

**Shashank Pandey · Mission Forge**